The entrances to the wormhole are called “mouths.” You are sitting in a sidewalk cafe alongside the Dublin mouth. I am standing in the desert beside the California mouth. Both mouths look rather like crystal balls. When I look into my California mouth, I see a distorted image of Grafton Street, Dublin (Figure 14.6). That image is brought to me by light that travels through the wormhole from Dublin to California, rather like light traveling through an optical fiber. When you look into your Dublin mouth, you see a distorted image of Joshua trees (cactus trees) in the California desert.

Can Wormholes Exist Naturally, as Astrophysical Objects?



In *Interstellar*, Cooper says, “A wormhole isn’t a naturally occurring phenomenon.” I agree with him completely! If traversable wormholes are allowed by the laws of physics, I think it extremely unlikely they can exist naturally, in the real universe. I must confess, though, that this is little more than a speculation, not even an educated guess. Maybe a highly educated speculation, but speculation nonetheless, so I labeled this section  .

Why am I so pessimistic about natural wormholes?

We see no objects in our universe that could become wormholes as they age. By contrast, astronomers see huge numbers of massive stars that will collapse to form black holes when they have exhausted their nuclear fuel.

On the other hand, there is reason to hope that wormholes *do* exist naturally on submicroscopic scales in the form of “quantum foam” (Figure 14.7). This foam is a hypothesized network of wormholes that is continually fluctuating in and out of existence in a manner governed by